Mode 9 — Square-BFP (centered / opt)

DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered add-and-square

Step 1 — DP8 (16×16)

$$\mathrm{DP8}\,(16\times16) = \sum_{i=0}^{7} a_i\, b_i$$

Step 2 — DP8 (16×8)

$$\mathrm{DP8}\,(16\times16) = 2^8 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hi}}_{\mathrm{DP8}\,(16\times8)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lo}}_{\mathrm{DP8}\,(16\times8)}$$

Step 3 — DP8 (16×4)

$$\mathrm{DP8}\,(16\times16) = 2^{12} \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hh}}_{\mathrm{DP8}\,(16\times4)} + 2^8 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{hl}}_{\mathrm{DP8}\,(16\times4)} + 2^4 \underbrace{\sum_{i=0}^{7} a_i\, b_i^{lh}}_{\mathrm{DP8}\,(16\times4)} + \underbrace{\sum_{i=0}^{7} a_i\, b_i^{ll}}_{\mathrm{DP8}\,(16\times4)}$$

Step 4 — DP8 (8×4)

$$\mathrm{DP8}\,(16\times16) = 2^{20} \underbrace{\sum_{i=0}^{7} a_i^{hi} b_i^{hh}}_{\mathrm{DP8}\,(8\times4)} + 2^{16} \underbrace{\sum_{i=0}^{7} a_i^{hi} b_i^{hl}}_{\mathrm{DP8}\,(8\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_i^{lo} b_i^{hh}}_{\mathrm{DP8}\,(8\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_i^{hi} b_i^{lh}}_{\mathrm{DP8}\,(8\times4)}$$

$$+ 2^{8} \underbrace{\sum_{i=0}^{7} a_i^{lo} b_i^{hl}}_{\mathrm{DP8}\,(8\times4)} + 2^{8} \underbrace{\sum_{i=0}^{7} a_i^{hi} b_i^{ll}}_{\mathrm{DP8}\,(8\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_i^{lo} b_i^{lh}}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i^{lo} b_i^{ll}}_{\mathrm{DP8}\,(8\times4)}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E; parallel matmul: own E_a, E_b)

$$\big(P_j,\ E\big) \ \text{ per DP8}\,(8\times4)\text{ above,} \qquad E = E_a + E_b \ \longrightarrow \ \big(M,\ E\big)$$

$\delta \equiv 0$: M is the Step-4 sum unchanged; $\mathrm{DP8}\,(16\times16) = M \cdot 2^E$.

Step 6 — DP8 (4×4)

$$M = 2^{24} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi} b_i^{hh}}_{\mathrm{DP8}\,(4\times4)} + 2^{20} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi} b_i^{hh}}_{\mathrm{DP8}\,(4\times4)} + 2^{20} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi} b_i^{hl}}_{\mathrm{DP8}\,(4\times4)} + 2^{16} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi} b_i^{hl}}_{\mathrm{DP8}\,(4\times4)}$$

$$+ 2^{16} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo} b_i^{hh}}_{\mathrm{DP8}\,(4\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo} b_i^{hh}}_{\mathrm{DP8}\,(4\times4)} + 2^{16} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi} b_i^{lh}}_{\mathrm{DP8}\,(4\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi} b_i^{lh}}_{\mathrm{DP8}\,(4\times4)}$$

$$+ 2^{12} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo} b_i^{hl}}_{\mathrm{DP8}\,(4\times4)} + 2^{8} \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo} b_i^{hl}}_{\mathrm{DP8}\,(4\times4)} + 2^{12} \underbrace{\sum_{i=0}^{7} a_{H,i}^{hi} b_i^{ll}}_{\mathrm{DP8}\,(4\times4)} + 2^{8} \underbrace{\sum_{i=0}^{7} a_{L,i}^{hi} b_i^{ll}}_{\mathrm{DP8}\,(4\times4)}$$

$$+ 2^{8} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo} b_i^{lh}}_{\mathrm{DP8}\,(4\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo} b_i^{lh}}_{\mathrm{DP8}\,(4\times4)} + 2^{4} \underbrace{\sum_{i=0}^{7} a_{H,i}^{lo} b_i^{ll}}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=0}^{7} a_{L,i}^{lo} b_i^{ll}}_{\mathrm{DP8}\,(4\times4)}$$

Step 7 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$M = 2^{24}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a^{hi}_{H,i}+b^{hh}_i)^2 - \sum_{i=0}^{7}(a^{hi}_{H,i})^2 - \sum_{i=0}^{7}(b^{hh}_i)^2\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{20}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+b^{hh}_i\big)^2 - \sum_{i=0}^{7}(a^{hi}_{L,i}-8)^2 - \sum_{i=0}^{7}(b^{hh}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{20}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{hl}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 - \sum_{i=0}^{7}(b^{hl}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{16}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{hl}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{hl}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{16}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+b^{hh}_i\big)^2 - \sum_{i=0}^{7}(a^{lo}_{H,i}-8)^2 - \sum_{i=0}^{7}(b^{hh}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{12}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+b^{hh}_i\big)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i}-8)^2 - \sum_{i=0}^{7}(b^{hh}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{16}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{lh}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 - \sum_{i=0}^{7}(b^{lh}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{12}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{lh}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{lh}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{12}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{hl}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 - \sum_{i=0}^{7}(b^{hl}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{8}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{hl}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{hl}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{12}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{ll}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 - \sum_{i=0}^{7}(b^{ll}_i-8)^2 + 512\Big)}_{\text{DP8 (4×4)}}$$

$$+\,2^{8}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{ll}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{ll}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{lh}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 - \sum_{i=0}^{7}(b^{lh}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{lh}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{lh}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{ll}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 - \sum_{i=0}^{7}(b^{ll}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{ll}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{ll}_i-16)^2 + 2048\Big)}_{\text{DP8 (4×4)}}$$

Step 8 — All components

$$\begin{aligned}
\mathrm{PE} = {} & 2^{24}\sum_{i=0}^{7}(a^{hi}_{H,i}+b^{hh}_i)^2 + 2^{20}\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+b^{hh}_i\big)^2 + 2^{20}\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{hl}_i-8)\big)^2 + 2^{16}\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{hl}_i-8)\big)^2 \\
& + 2^{16}\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+b^{hh}_i\big)^2 + 2^{12}\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+b^{hh}_i\big)^2 + 2^{16}\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{lh}_i-8)\big)^2 + 2^{12}\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{lh}_i-8)\big)^2 \\
& + 2^{12}\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{hl}_i-8)\big)^2 + 2^{8}\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{hl}_i-8)\big)^2 + 2^{12}\sum_{i=0}^{7}\big(a^{hi}_{H,i}+(b^{ll}_i-8)\big)^2 + 2^{8}\sum_{i=0}^{7}\big((a^{hi}_{L,i}-8)+(b^{ll}_i-8)\big)^2 \\
& + 2^{8}\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{lh}_i-8)\big)^2 + 2^{4}\sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{lh}_i-8)\big)^2 + 2^{4}\sum_{i=0}^{7}\big((a^{lo}_{H,i}-8)+(b^{ll}_i-8)\big)^2 + \sum_{i=0}^{7}\big((a^{lo}_{L,i}-8)+(b^{ll}_i-8)\big)^2
\end{aligned}$$

$$\begin{aligned}
\alpha = {} & 2^{24}\sum_{i=0}^{7}(a^{hi}_{H,i})^2 + 2^{20}\sum_{i=0}^{7}(a^{hi}_{L,i}-8)^2 + 2^{20}\sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 + 2^{16}\sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 \\
& + 2^{16}\sum_{i=0}^{7}(a^{lo}_{H,i}-8)^2 + 2^{12}\sum_{i=0}^{7}(a^{lo}_{L,i}-8)^2 + 2^{16}\sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 \\
& + 2^{12}\sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 + 2^{8}\sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2 + 2^{12}\sum_{i=0}^{7}(a^{hi}_{H,i}-8)^2 + 2^{8}\sum_{i=0}^{7}(a^{hi}_{L,i}-16)^2 \\
& + 2^{8}\sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2 + 2^{4}\sum_{i=0}^{7}(a^{lo}_{H,i}-16)^2 + \sum_{i=0}^{7}(a^{lo}_{L,i}-16)^2
\end{aligned}$$

$$\begin{aligned}
\beta = {} & 2^{24}\sum_{i=0}^{7}(b^{hh}_i)^2 + 2^{20}\sum_{i=0}^{7}(b^{hh}_i-8)^2 + 2^{20}\sum_{i=0}^{7}(b^{hl}_i-8)^2 + 2^{16}\sum_{i=0}^{7}(b^{hl}_i-16)^2 \\
& + 2^{16}\sum_{i=0}^{7}(b^{hh}_i-8)^2 + 2^{12}\sum_{i=0}^{7}(b^{hh}_i-8)^2 + 2^{16}\sum_{i=0}^{7}(b^{lh}_i-8)^2 + 2^{12}\sum_{i=0}^{7}(b^{lh}_i-16)^2 \\
& + 2^{12}\sum_{i=0}^{7}(b^{hl}_i-16)^2 + 2^{8}\sum_{i=0}^{7}(b^{hl}_i-16)^2 + 2^{12}\sum_{i=0}^{7}(b^{ll}_i-8)^2 + 2^{8}\sum_{i=0}^{7}(b^{ll}_i-16)^2 \\
& + 2^{8}\sum_{i=0}^{7}(b^{lh}_i-16)^2 + 2^{4}\sum_{i=0}^{7}(b^{lh}_i-16)^2 + 2^{4}\sum_{i=0}^{7}(b^{ll}_i-16)^2 + \sum_{i=0}^{7}(b^{ll}_i-16)^2
\end{aligned}$$

$$\begin{aligned}
C = {} & 0 + 536870912 + 536870912 + 134217728 + 33554432 + 2097152 + 33554432 + 8388608 \\
& + 8388608 + 524288 + 2097152 + 524288 + 524288 + 32768 + 32768 + 2048 = 1297680384
\end{aligned}$$

$$M = \tfrac{1}{2}\big(\mathrm{PE} - \alpha - \beta + C\big), \qquad E = E_a + E_b$$